5	**(a)**		$(\mathbf{W} =)\ -2.5g\mathbf{j}$ N $\quad (= -24.5\mathbf{j}$ N)	**B1** **[1]**	**1.2**	Condone missing units. Must be negative. Allow any correct vector notation	
5	**(b)**		Newton's second law $-24.5\mathbf{j} + (3\mathbf{i} - 2\mathbf{j}) + (-\mathbf{i} + 18\mathbf{j}) = 2.5\mathbf{a}$ $\mathbf{a} = \frac{1}{2.5}(2\mathbf{i} - 8.5\mathbf{j}) = (0.8\mathbf{i} - 3.4\mathbf{j})$ m s^{-2}	**M1** **A1** **[2]**	**1.1a** **1.1**	Attempt to use N2L with $m = 2.5$ Condone missing weight or an incorrect weight vector Cao – do not award if the magnitude of acceleration given as acceleration	Must be sum of forces = $2.5a$